

Comparing Performance Patterns in Tested vs. Untested Powerlifting Competition Entries

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Motivation

In gym culture, few questions spark more debate than "Are you natty or not?" Here, "natty" refers to people who don't use performance-enhancing drugs (PEDs). Competitive powerlifting lets me study this debate with real data, because some competition entries count as drug-tested and others do not. The OpenPowerlifting public dataset has millions of records covering both types of entries.

I chose this topic because it connects a widely discussed question in strength sports with a dataset large enough for a one-semester study. The project is also analytically interesting: if tested and untested entries represent different competition contexts, similar athletes may show different performance patterns across them. I'm not trying to identify whether individual athletes use PEDs. Instead, I use the OpenPowerlifting "Tested" field as a row-level competition-entry label and ask whether lifting performance differs systematically between tested and untested entries. I expect this project to show whether those performance gaps appear consistently and which athlete groups show the largest differences.

Dataset

Source: OpenPowerlifting, a public, community-maintained archive of powerlifting competition data.

Size: Over 3.8 million competition records spanning many meets, years, and countries. I'll analyze a filtered subset so the work remains feasible in a notebook-based workflow.

Key Features: The dataset captures who the athlete is (sex, age, bodyweight, weight class), what they lifted (best squat, bench, deadlift, and Total (kg)), and where and how they competed (federation, equipment type, tested status). The Dynamic Objective Team Scoring (DOTS) formula provides a standardized scoring field for cross-weight-class comparisons.

Preprocessing: Older meets have more missing fields and inconsistent types, so I'll filter down to adult open athletes with valid Total (kg) values and standardize categorical fields.

Potential Challenges: I expect to deal with uneven numbers of tested and untested records and missing data in older meets. Because the "Tested" field is a row-level entry label rather than proof of a drug test, I'll interpret the results as differences between competition contexts.

Research Questions

RQ1 - Natty or Not? (Main Performance Comparison): Do untested entries tend to show higher Total (kg) and DOTS scores? I'll group athletes by sex, weight class, and equipment for apples-to-apples comparisons. *H0: Untested entries do not have higher mean DOTS/Total than tested entries within the same subgroup. Ha: Untested entries have higher mean DOTS/Total (one-sided).*

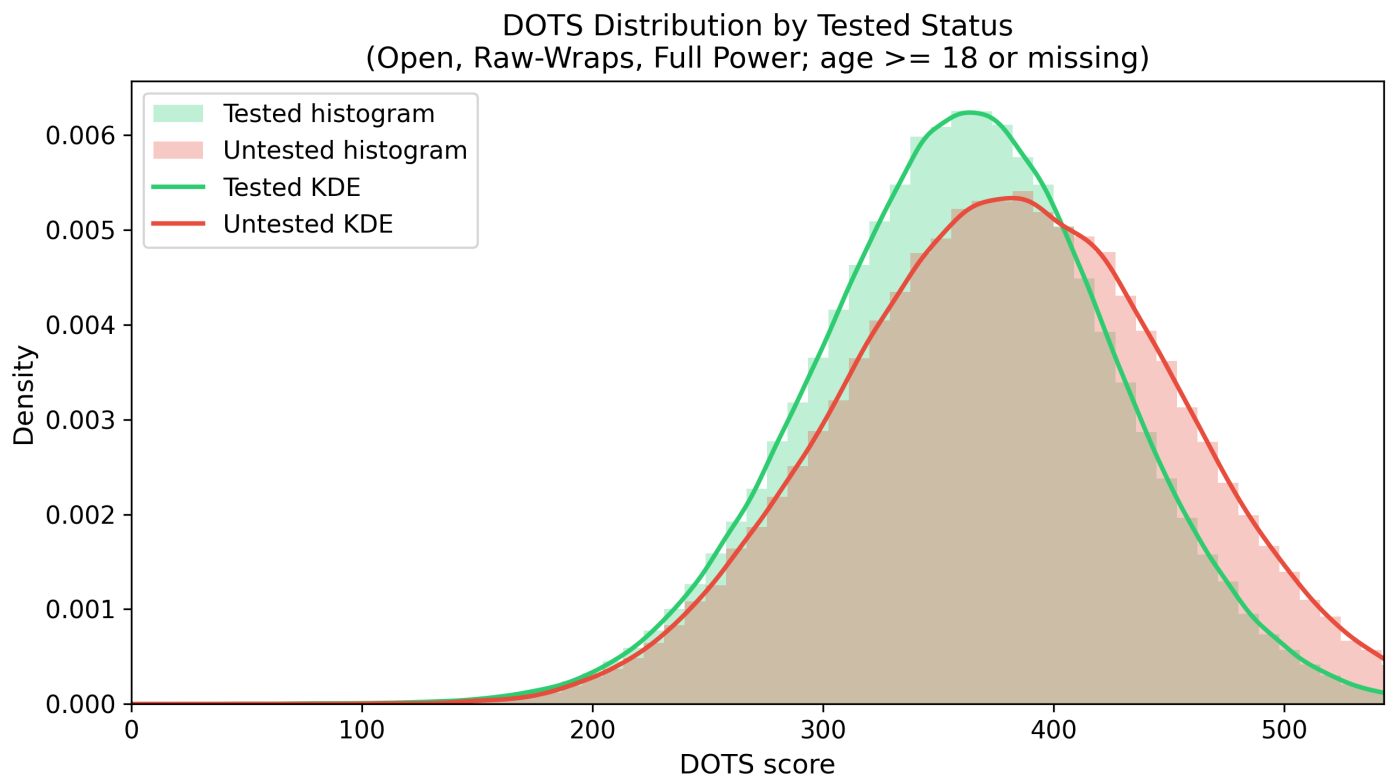


Figure 1. A first look at how DOTS scores compare between tested and untested entries before breaking things down by sex, weight class, or equipment.

RQ2 - Testing, Testing... Anyone Still Here? (Trend Analysis): Is tested status associated with DOTS-score bin, separately by sex? I'll divide entries into evenly spaced DOTS bins and compare the tested-entry share across bins for men and women. *H0: DOTS-score bin and tested status are independent. Ha: DOTS-score bin and tested status are associated within each sex.*

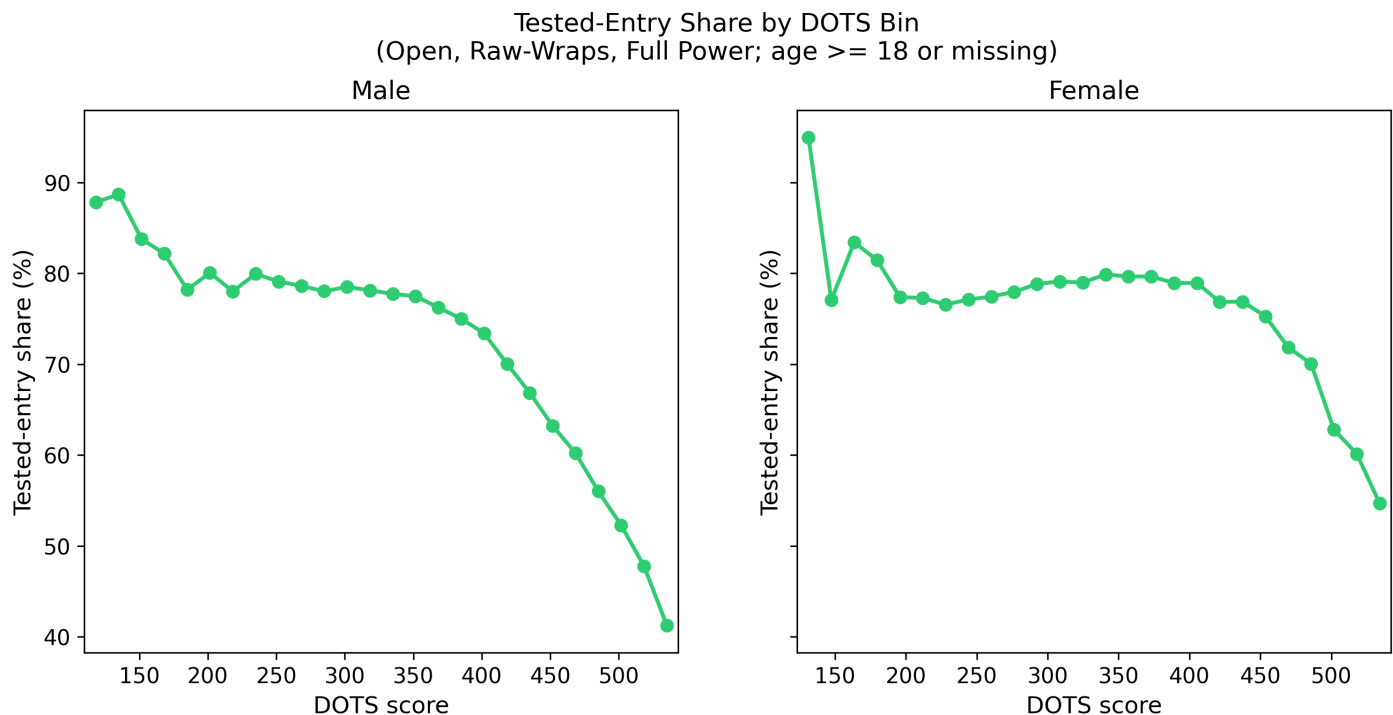


Figure 2. A preliminary look at how the share of tested entries changes as DOTS score increases, shown separately for men and women. The apparent trend motivates the formal analysis in RQ2.

RQ3 - Age Is Just a Number... Or Is It? (Age Pattern Comparison): Do untested entries tend to show higher mean Total (kg) across age bins, and how does that gap look at different ages? I'll compare the age-profile lines and the tested-vs-unttested gap within each age bin and use bootstrap intervals to check whether the non-tested-minus-tested gap in each age bin is clearly above zero, treating the direction and size of the gap as the main evidence.

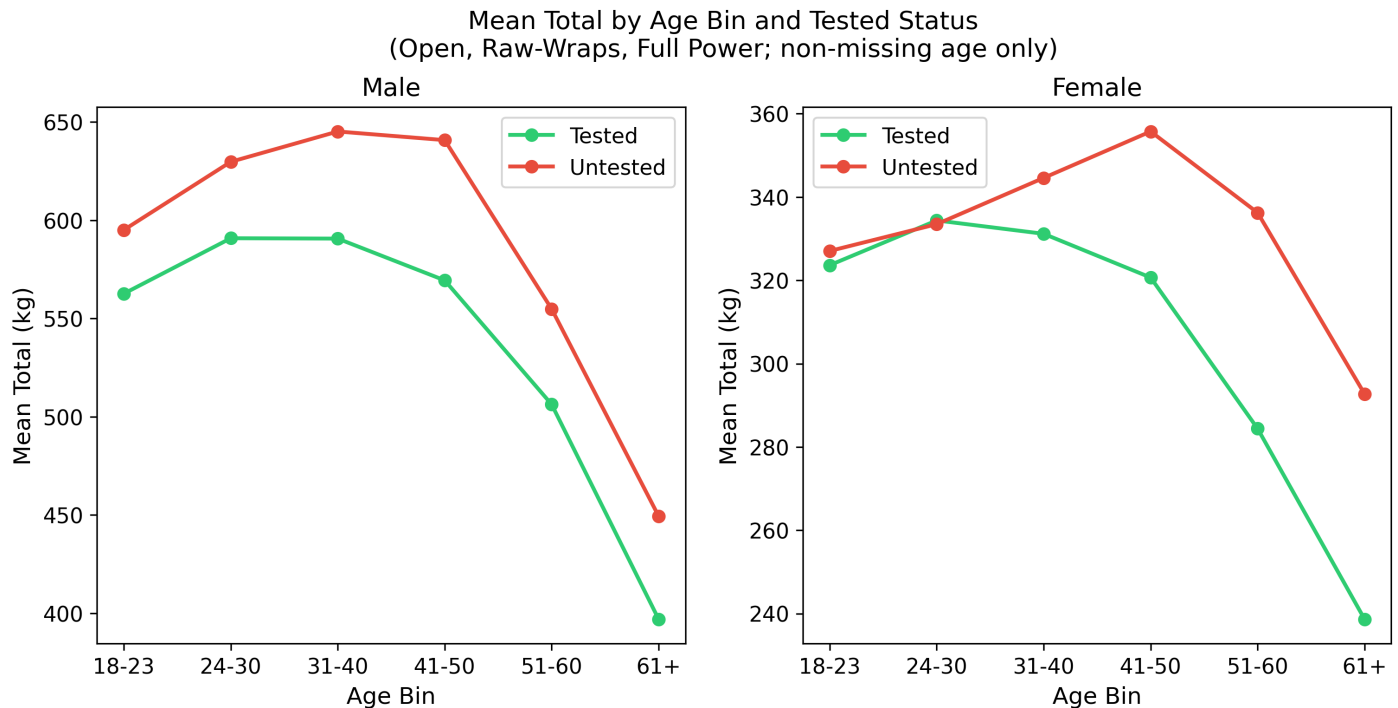


Figure 3. An early look at how mean Total (kg) compares across age groups for tested and untested entries, split by sex. These patterns motivate RQ3.

Methods / Approach

RQ1 and RQ2 each have a formal null and alternative hypothesis, tested using methods from the course's hypothesis-testing guidance. RQ3 takes a descriptive approach, using bootstrap confidence intervals to assess the size and direction of performance gaps across age bins. The core pipeline below covers the data preparation, visualization, and inferential steps for each part of the analysis.

Core Pipeline:

- 1. Data cleaning and preparation:** I'll load the data, inspect and clean types and missing values, and filter to the analysis subset.
- 2. Feature engineering and grouping:** I'll create age bins and build grouped summaries to compare tested and untested performance patterns.
- 3. Descriptive visualization:** I'll use histograms, bar charts, line plots, and KDE plots to compare tested and untested distributions across sex, equipment, and age groups (KDE bandwidth via Silverman's rule).
- 4. RQ1 main performance comparison:** Within each athlete subgroup (sex, weight class, equipment), I'll compare Total (kg) and DOTS between tested and untested entries using a one-sided Welch's t-test and a one-sided permutation test, with bootstrap CIs for the size of each gap.
- 5. RQ2 trend analysis:** Separately by sex, I'll cross-tabulate DOTS-score bins by tested status, visualize the tested-entry share by bin, and use a chi-squared test of independence to assess whether tested status varies across bins. If the tested share rises or falls across bins, I'll describe the pattern but won't test it as a separate hypothesis.

6. **RQ3 age-pattern comparison:** I'll compute mean Total (kg) by age bin, tested status, and sex; visualize the two age profiles; and use bootstrap 95% confidence intervals for the tested-vs-unttested gap within each age bin. I will treat differences in peak age or widening/narrowing gaps as descriptive findings rather than separate formal hypothesis tests.

References

OpenPowerlifting. *OpenPowerlifting Dataset (Revision: dfb517af)*. Retrieved on March 14, 2026 from <https://openpowerlifting.gitlab.io/opl-csv/bulk-csv.html>. Licensed under Creative Commons Zero (CC0).

Kopayev, O., Onyshchenko, B., & Stetsenko, A. (2020). Report: Evaluation of Wilks, Wilks-2, DOTS, IPF and Goodlift Formulas for Calculating Relative Scores in IPF Powerlifting Competitions. International Powerlifting Federation. https://www.powerlifting.sport/fileadmin/ipf/data/ipf-formula/Models_Evaluation-I-2020.pdf